

# **EXPERIMENT - 8**

Name : Yashwardhan Chourasia

PRN : 25070521167

## **8.1.2. Prime Numbers in a Given Range**

### **A) ALGORITHM**

Step 1: Start

Step 2: Read the starting number (start).

Step 3: Read the ending number (stop).

Step 4: For num = start to stop

    If num > 1

        Set flag = True

        For i = 2 to  $\sqrt{\text{num}}$

            If num % i == 0

                Set flag = False

                Break

        End For

        If flag == True

            Print num

    End If

Step 5: Stop

### **B) PYTHON CODE**

```
start = int(input())
```

```
stop = int(input())
```

```
f=False
```

```
for num in range(start, stop + 1):
```

```
    if num > 1:
```

```
        flag = True
```

```
    for i in range(2, int(num ** 0.5) + 1):
```

```
        if num % i == 0:
```

```
            flag = False
```

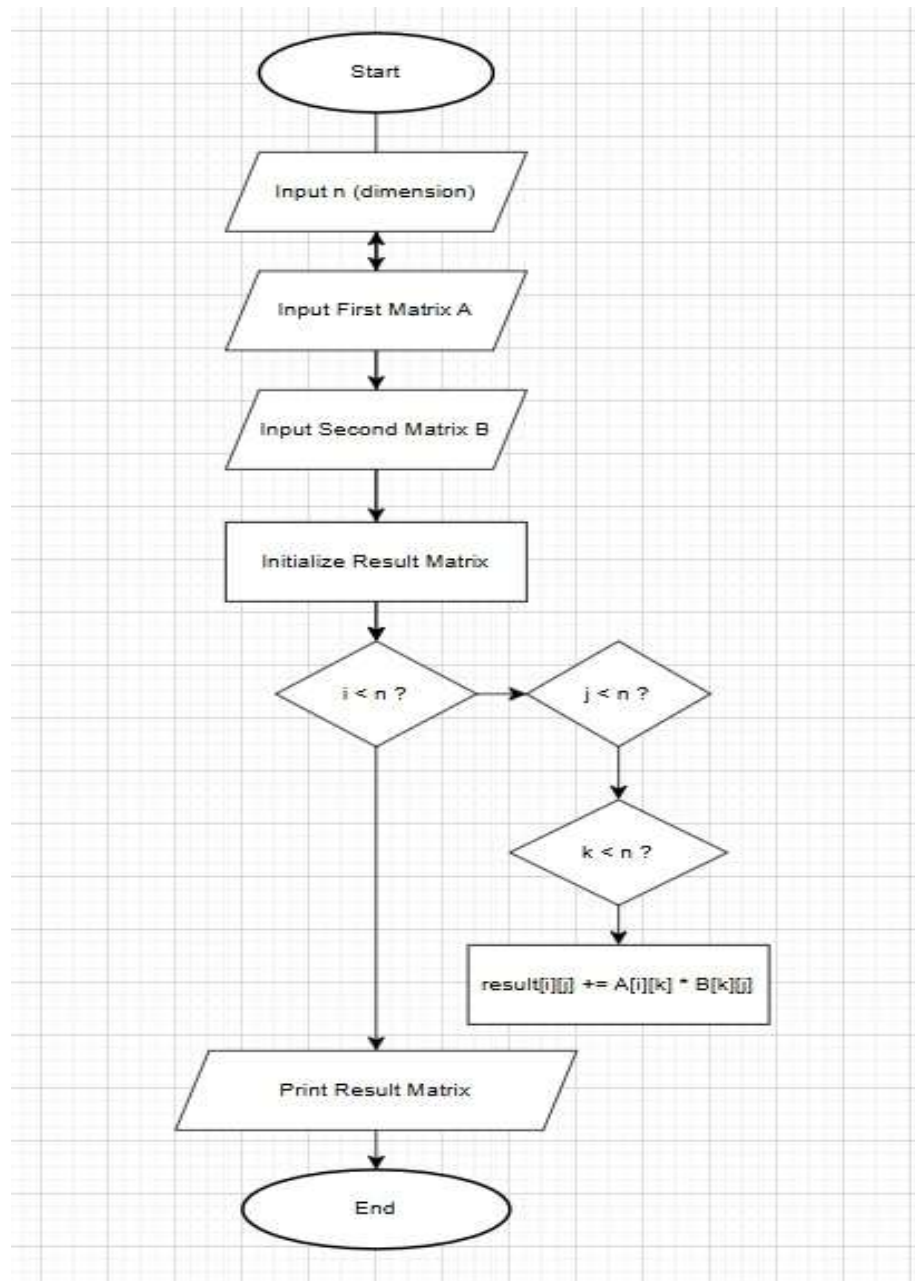
```
            break
```

```
    if flag
```

```
        print(num)        f=True
```

```
if not f:print("No primes")
```

### C) FLOWCHART



D) EXECUTION

CODETANTRA

Home

8.1.2. Prime Numbers In a Given Range

00:13

Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).

Input Format:

- First line contains an integer representing the start of the range.
- Second line contains an integer representing the end of the range.

Output Format:

- Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a separate line.
- If no prime numbers exist in the range, print: "No primes".

Sample Test Cases

Logout

yashwardhan.chourasia.batch2025@sitnagpur.siu.edu.in

Support

primeNu...

1 start = int(input())

2 stop = int(input())

3 f=False

4 for num in range(start, stop + 1):

5 if num > 1:

6 flag = True

7

8 for i in range(2, int(num \*\* 0.5) + 1):

9 if num % i == 0:

10 flag = False

11 break

Debugger

Submit

Average time 0.010 s

Maximum time 0.014 s

9.56 ms

14.00 ms

5 out of 5 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 14 ms

Debug

Expected output

Actual output

10 10

20 20

11 11

13 13

17 17

19 19

Terminal

Test cases

Prev

Reset

Submit

Next